Lecture plan · for the lecturer

Adversarial examples

AI security · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- Explain why adversarial examples exist.
- Compute and run FGSM.
- Evaluate defences honestly.

Activity: Attack, defend, attack again

1. Warm-up (5 min): ماذا يرى النموذج؟
2. Attack (15 min): Pairs compute ɛllwll for a paper classifier and flip its prediction.
3. Defend (15 min): Each pair proposes a defence and states its threat model.
4. Break (20 min): Swap and design an attack that knows the other pair's defence.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which defences survived?
- What did the threat model leave out?
- How would you report robustness?

Exit ticket

- Name one claim you will no longer accept without an adaptive attack.

Try it in your next lecture
Give pairs a three-feature linear classifier on paper; one attacks it with ε = 0.1, the other proposes a defence, then they swap and attack the defence.

Seminar sheet · for students

Adversarial examples

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Name _____ Class _____ Date _____

1 **A defence stops FGSM. What test comes next?**

◯ Run FGSM again with a new seed ◯ An adaptive attack designed for the defence

◯ Report clean accuracy only

2 **Bound the change.**

A linear model has w = (1, -2, 4). $\Sigma|w_i|$ = _____, so with ε = 0.05 FGSM can change the score by _____.

Word bank: 7 · 3 · 0.15 · 0.35

3 **Which belong in a sound robustness evaluation? Choose all that apply.**

☐ A stated threat model and ε ☐ An adaptive attack designed for the defence

☐ Iterative attacks such as PGD, not only FGSM ☐ Reporting only clean accuracy

4 **If FGSM fails against a defence, the model is robust within that ε.**

◯ True ◯ False

5 **Match each idea to its source.**

Linearity explains adversarial examples	• • • •	Eykholt et al., 2018
Adversarial training as min-max	• • • •	Athalye et al., 2018
Obfuscated gradients break defences	• • • •	Madry et al., 2018
Physical stickers on stop signs	• • • •	Goodfellow et al., 2015

✎ **When your students claim a model is secure, what evidence do you currently accept?**